

Microwave Tumor Ablation: Mechanism of Action, Clinical Results, and Devices

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Microwave ablation uses dielectric hysteresis to produce direct volume heating of tissue. Microwaves are capable of propagating through many tissue types, even those with high impedance such as lung or bone, with less susceptibility to “heat-sink” effects along vessels. Microwaves are highly conducive to the use of multiple applicators, showing the synergy seen with other energies, but also the potential capability for phasing of the electromagnetic field. As a result, larger, more customizable ablation zones may be created in less time. Although multiple microwave ablation systems are currently available, further study and continued development are needed.

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Abbreviation: RF = radiofrequency

HOW MICROWAVE ABLATION WORKS

Tissue Heating

MICROWAVE ablation uses dielectric hysteresis to produce heat. Tissue destruction occurs when tissues are heated to lethal temperatures from an applied electromagnetic field, typically at 900–2,500 MHz. Polar molecules in tissue (primarily H₂O) are forced to continuously realign with the oscillating electric field, increasing their kinetic energy and, hence, the temperature of the tissue (Fig 1). Tissues with a high percentage of water (eg, solid organs and tumors) are most conducive to this type of heating (1–8).

Microwave energy radiates into the

tissue through an interstitial antenna that functions to couple energy from the generator power source to tissue. As a result of the radiating nature of the antenna, direct heating occurs in a volume of tissue around the antenna. This mechanism of heating differs substantially from radiofrequency (RF) ablation, which creates heat via resistive heating when electrical current passes through the ionic tissue medium. RF heating requires an electrically conductive path, is limited in areas of low electrical conductivity, and results in heating of tissues only immediately adjacent to the electrode (3,9,10). Microwaves are capable of propagating through and effectively heating many types of tissue, even those with low electrical conductivity, high impedance, or low thermal conductivity. For example, bone and lung are two tissue types that have been associated with suboptimal outcomes or local progression with RF ablation as a result of high baseline impedance (2,11–17). Unlike RF and laser, microwaves can readily penetrate through the charred or desiccated tissues that tend to build up around all hyperthermic ablation applicators, resulting in limited power delivery for energy systems other than microwave systems (18). Relative permittivity (ie, dielectric constant) is a measure of how well a material or tissue will accept an electric field and is another tissue property that

can impact how well energy will propagate through tissue (3).

Multiple microwave antennas can be powered simultaneously to take advantage of thermal synergy when placed in close proximity, or widely spaced to ablate several tumors simultaneously (Fig 2) (19–22). Multiapplicator ablation is possible with other power sources, but unlike RF, microwave is able to be powered continuously without switching from one electrode to another during electrode activation. Also unique to microwave ablation is the ability for antennas to be positioned and phased to exploit overlap of the electromagnetic field (19–27). When phased constructively, heating increases proportional to the square of the number of antennas, allowing more efficient heating and generation of higher temperatures compared with a single antenna (24,25,27). This increase in heating is in addition to the thermal synergy seen with other currently available multiapplicator ablation technologies. In the future, phasing may be able to be exploited in a way that will allow larger and more customizable ablation zones than is currently possible.

Components of a Microwave System

The basic microwave system consists of three components: a generator, a power distribution system, and anten-

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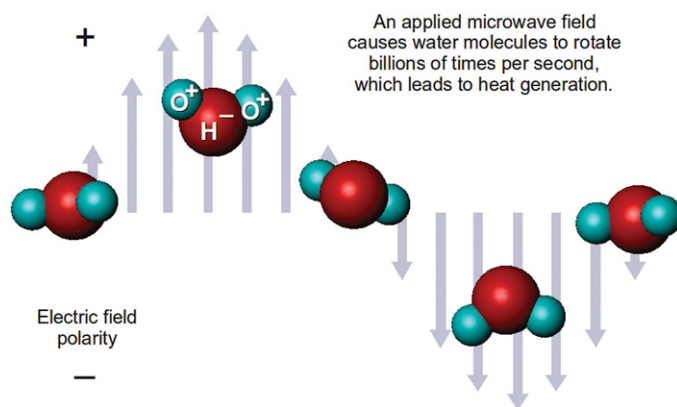


Figure 1. Microwave heating physics: schematic demonstrates how an alternating electromagnetic field causes polar molecules to continuously realign, producing kinetic energy and, in turn, heat. Figure reprinted from Brace (1) with permission from Elsevier.

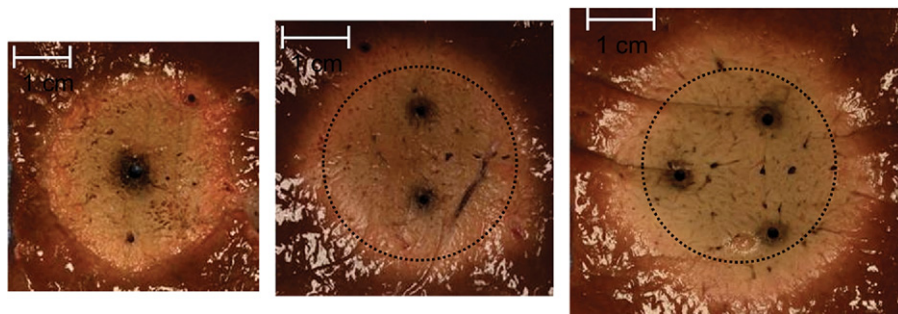


Figure 2. Images obtained during ex vivo ablations with one, two, and three antennas, respectively, powered by a 2.45-GHz generator delivering a total of 90 W (1×90 W, 2×45 W, or 3×30 W) for 5 minutes, demonstrate the effects of thermal synergy. The dotted circle represents the ablation zone created with a single antenna superimposed on the multiplier ablations.

nas. As a result of shaft heating caused by reflected power, a cooling system is a crucial component of most microwave antennas, and most currently available microwave systems in the United States use an antenna cooling mechanism.

Power is generated with magnetron or solid-state sources. Microwave generator output can be controlled relatively independently of the tissue type; the impedance spikes or reduced power output characteristic of RF ablation in high-impedance tissues are not encountered in microwave ablation. **Generator frequencies are generally 915 MHz or 2.45 GHz, as allowed by the Federal Communications Commission (1,6,22,28–34).** To date, there are limited data to suggest that any given frequency is more effective for microwave ablation procedures; however, a single preclinical study directly comparing 915 MHz and 2.45 GHz systems (35) has suggested that a 915-MHz generator frequency used in

combination with a cooled-shaft antenna may generate larger ablation zones than a similar 2.45-GHz system. The 915-MHz ablation zones were also very long and therefore may be limited in the anatomic areas in which they can be used. Additional study is needed to identify whether these results are reproducible and/or tissue-dependent.

Distribution of electromagnetic energy from the generator to the antenna is most commonly accomplished through a coaxial transmission line. Coaxial cables have excellent propagation characteristics, but as cable diameter decreases, power loss (and associated cable heating) increases. Therefore, there is a limit to how small and flexible the cables can be without resulting in dangerous cable heating (1).

Microwave antennas are the final and most critical component of the system, functioning to transfer energy into tissue. The active heating zone and power coupling efficiency of an antenna

is determined by its geometry. Most microwave ablation antenna designs are straight and needle-like (7,36,37). Common designs include monopole, dipole, triaxial, choked, or slotted antennas (38–44).

Microwave antenna design is a balance of power efficiency, tissue heating pattern, and antenna diameter with design tradeoffs necessary to produce a specific desired result. As antennas are generally constructed from coaxial cable, smaller-diameter antennas can have trouble handling higher power levels without unwanted thermal damage to tissues around the proximal antenna shaft (Fig 3) (6,17,28,45,46). Cooling jackets and antenna shaft cooling systems have been shown to reduce this heating, eliminate skin burns, and increase power handling of smaller-diameter antennas (31,47).

Circulation of chilled saline solution or water is the most commonly used method for cooling the antenna shaft, and the addition of active cooling has enabled delivery of higher powers for longer times, and in turn, production of larger ablation zones (31). Another strategy for antenna cooling is the use of compressed gas, which is used by at least one system (Certus 140; NeuWave Medical, Madison, Wisconsin). Rapid decompression of carbon dioxide gas causes the Joule-Thomson phenomenon to occur at the probe tip, with gas venting up the shaft and along feed lines. The high cooling capacity of this system allows the use of high-power generators (140 W) while maintaining small shaft diameters (17-gauge).

Ultimately, the ablation zone size and shape produced by any antenna in live tissue depends on the antenna design, tissue type (taking into account the changes in the tissue properties during the ablation), thermal conduction from the active heating zone, and thermal “sinks” caused by nearby structures such as blood vessels. The interplay among these factors is complex, and direct comparisons among the various system designs have not yet been accomplished.

ADVANTAGES OF MICROWAVE ABLATION

Global

Microwave has many theoretical advantages over current technologies: microwave energy has the po-

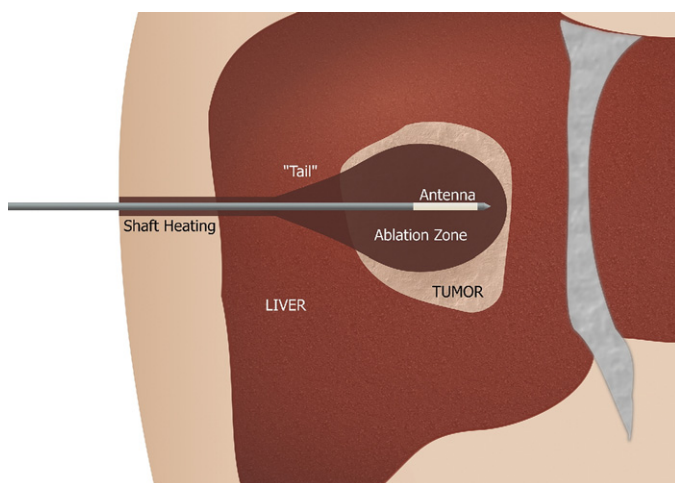


Figure 3. Shaft heating: because of the significant shaft heating that can occur with microwaves, a robust shaft cooling mechanism is required to minimize thermal damage to the subcutaneous tissues and the skin, especially with the development of higher-power systems.

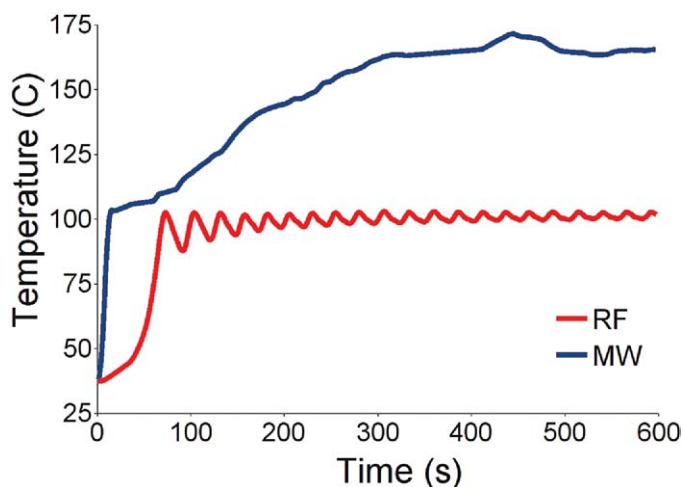


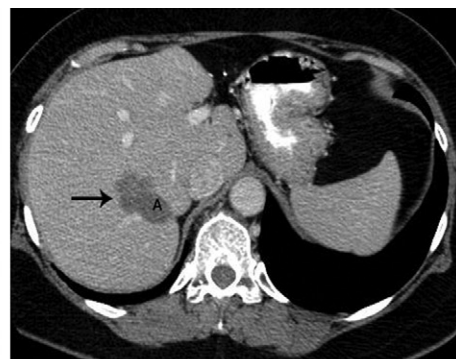
Figure 4. Microwave versus RF temperatures in porcine kidney in vivo. Data collected with a fiberoptic sensor 5 mm away from an RF electrode (200 W, impedance switching; Cool-tip) or microwave antenna (triaxial, 50 W, 2.45 GHz) in normal porcine kidney in vivo show higher temperatures (well greater than 100°C) over time around the microwave antenna (65).



a.



b.



c.

Figure 5. Recurrence along a hepatic vein after RF ablation. Pretreatment CT image (**a**) demonstrates a small low-attenuation lesion located between hepatic venous branches (arrow), which suggested metastatic disease in this patient with colorectal cancer. Ultrasound image during treatment with RF ablation (**b**) demonstrates gas encompassing the small lesion (arrow). Follow-up CT (**c**) demonstrates a low-attenuation ablation zone (A) with local tumor progression along the margin abutting the hepatic vein (arrow).

tential to produce faster heating over a larger volume of tissue with less susceptibility to heat-sink effects; can be effective in tissues with high impedance such as lung or charred, desiccated tissue; is capable of generating very high temperatures, often in excess of 100°C (**Fig 4**); is highly conducive to the use of multiple applicators; and does not require grounding pads or other ancillary components (19,20,22). Microwave also has certain organ-specific advantages, as detailed in the subsequent paragraphs.

Liver

The liver is a vascular solid organ with an abundance of large vessels creating the potential for heat-sink effects (**Fig 5**). Microwaves appear to be more apt to overcome perfusion and large heat sinks than other heat-based ablation modalities (5,10,48–50). Microwave energy has been shown to ablate tissue up to and around large hepatic vessels as large as 10 mm and creates larger zones of ablation in high-perfusion areas (5,49,50). High perfusion rates in hepatic vessels greater than 3 mm limits

the effectiveness of RF ablation, and has been shown to be an independent predictor of incomplete tumor destruction (51).

The decreased susceptibility to vascular cooling has been studied and confirmed in preclinical studies. Awad et al (50) demonstrated large and consistent zones of ablation in shorter times than would normally be seen with RF ablation, and proximity to hepatic vasculature and inflow did not significantly change ablation zone size or shape with microwave ablation (**Table 1**) (5,6,50). In an in vivo porcine liver model, Brace et

Table 1
Preclinical Hepatic Microwave Trials (5,6,50)

Study, Year	Model	No. of Treatments	Treatment	System Parameters	Ablation Zone Size
Brace et al, 2007 (5)	In vivo porcine	58	OMWA	2.45 GHz, 68 W, 17-gauge, 2–12 min	Mean diameter, 2 min/2.3 cm, 6 min/2.6 cm, 12 min/2.9 cm
Awad et al, 2007 (50)	In vivo hepatic porcine	6	OMWA	2.45 GHz, 100 W, 5.7-mm antenna, 2–8 min	Mean volume, 2 min/33.5 cm ³ , 4 min/37.5 mm ³ , 8 min/92 cm ³
Hines-Peralta et al, 2006 (6)	Ex vivo bovine, in vivo porcine	165 (120/45)*	OMWA	2.45 GHz, 50–150 W, 5.7-mm antenna, 2–20 min	Ex vivo, 4.9 cm SAD; in vivo, 5.7 cm SAD

Note.—OMWA = open microwave ablation; SAD = short axis diameter.

* Ex vivo bovine/in vivo porcine.

al (5) created circular ablation zones with minimal effects related to even large intrahepatic vessels, suggesting that there is minimal heat-sink effect near vessels (Table 1).

Clinically, even with early-generation systems, microwave ablation has already been shown in several studies to have equal effectiveness, safety, and survival with shorter ablation times compared with RF ablation for the treatment of small hepatocellular carcinomas (8,46,52). Dong et al (52) described 234 patients who underwent percutaneous microwave ablation and demonstrated favorable survival without severe complications (Table 2) (8,30,36,46,52–56,64–66). Lu et al (8) retrospectively compared 102 patients who underwent treatment with microwave or RF ablation with no significant difference in survival or complication rates between the two groups (Table 2).

More recent studies with newer microwave systems have again demonstrated the efficacy of microwave ablation in the liver (8,53–57). Shiomi et al (53) compared percutaneous and thoracoscopy-assisted magnetic resonance imaging–guided microwave ablation in patients with hepatocellular carcinoma and metastatic disease, with 3-year survival rates of almost 90% in both groups for patients with hepatocellular carcinoma (median follow-up, 21 months; Table 2) (53). Ianitti et al (54) treated 87 patients with hepatocellular carcinoma and metastatic disease and found an overall survival rate of 47% (all tumor types) at 19 months (Table 2).

In addition, preclinical data have suggested that microwaves, particularly with the use of multiple applicators, may be effective in the treatment of larger tumors (> 3 cm) (19,20,28,36). Tu-

mors larger than 3 cm have historically been problematic for RF ablation, with a significantly increased risk of local tumor progression (58–60). However, the larger ablation zones possible with microwave ablation could potentially make these tumors more consistently treatable. For example, Brace et al (19) demonstrated ablation zones with mean diameters as large as 6.5 cm with the use of three 17-gauge microwave antennas spaced 3 cm apart in an in vivo porcine model. Strickland et al (28) used variable times and power outputs ranging from 36 to 200 W in an in vivo porcine liver model and demonstrated ablation zones ranging from 3 to 6 cm in diameter produced very rapidly, ie, within 3 minutes.

Early clinical data have supported the hypothesis that microwaves may be more effective against larger tumors (30,36,61–63). For example, Yu et al (61) treated four patients with hepatocellular carcinomas greater than 6 cm in diameter, and achieved complete ablation of three of the four lesions in two or three sessions (Table 2). Yin et al (64) treated patients with medium and large hepatic tumors. Although microwave showed a trend toward less local recurrence and larger ablation area than for a similarly sized hepatocellular carcinoma treated with RF ablation (96% for microwave, 90% for RF; $P = .288$), the differences were not statistically significant (Table 2) (64).

Similarly, early clinical studies (65, 66) have suggested that microwaves are effective in the treatment of colorectal hepatic metastatic disease, which requires a larger ablation margin and, hence, a larger ablation zone than for HCC. Shibata et al (66) prospectively randomized 30 patients with multiple

metastatic colorectal tumors to undergo microwave ablation or surgical resection and identified no significant difference in the 1-, 2-, and 3-year survival rates, with less blood loss in the microwave group (Table 2). Ogata et al (67) treated 102 unresectable colorectal metastatic lesions, with a high local control rate of 95% during a median follow-up of 33 months. However, new hepatic lesions or extrahepatic recurrence occurred in 78% of patients, and median survival time was 43 months.

Most authors report shorter ablation times in the liver with microwaves than with RF ablation, frequently less than 10 minutes, and many ranging from 2 to 5 minutes, depending on the number of applicators, lesion size, and power output. From a practical standpoint, decreased time needed for microwave ablation translates to more efficient use of equipment and personnel and decreased time for patients under general anesthesia, which is routinely used for ablation at our institution. However, just as with RF and cryoablation, use of general anesthesia may vary from site to site depending on physician preference. In addition, the speed of treatment gives microwaves an advantage for treating multiple lesions during one ablation session (63).

Kidney

The kidney is a highly vascular solid organ with high central perfusion rates creating significant heat-sink effects (68). Combined with concern for damaging the renal pelvis and ureter, large or centrally located tumors can be difficult to treat with RF ablation (69). The improved ability of microwaves to overcome the cooling effects of blood flow

Table 2
Clinical Hepatic Microwave Trials (8,30,36,46,52–56,64–66)

Study, Year	No. of Pts.	Treatment	System Parameters	Tumor Size
Yin et al, 2009 (64)	109	PMWA (<i>n</i> = 50), PRFA (<i>n</i> = 59)	MWA, 2.45 GHz, 60–70 W, 14–16-gauge, 5–20 min; RF, 290 KHz, 200 W, 14–17 gauge, 10–12 min	Mean 3.9 cm (20 tumors 5–7 cm)
Shiomi et al, 2008 (53)	161	TMWA (<i>n</i> = 69), PMWA (<i>n</i> = 73), OMWA, (<i>n</i> = 13), LMWA (<i>n</i> = 6)	2.45 GHz, repeated, 14-gauge, 60 W, 60-sec repeats	Mean: TMWA, 2.6 cm; PMWA, 2.4 cm
Ianitti et al, 2007 (54)	87	OMWA (<i>n</i> = 42), LMWA (<i>n</i> = 7), PMWA (<i>n</i> = 45)	915 MHz, 13-gauge, 45 W, 10 min	Mean 3.6 cm (range, 0.5–9.0 cm), 22 lesions > 4 cm
Yu et al, 2006 (36)	9	OMWA	915 MHz, 45–60 W, 13-gauge, 5–10 min	Mean 4.2 cm
Simon et al, 2006 (30)	10	OMWA	915 MHz, 45–60 W, 13 gauge × 3, 10 min	Mean 4.4 cm; CRC, HCC
Lu et al, 2005 (8)	102	PMWA (<i>n</i> = 49), PRFA (<i>n</i> = 53)	MWA, 2.45 GHz, 10–80 W, 16-gauge, 5-min repeats; RF, 290 KHz, 200 W, 14-gauge multitined electrode, ≥ 6 min	MWA, 2.5 cm ± 1.2; RF, 2.6 cm ± 1.2
Seki et al, 2005 (55)	68	LMWA	2.45 GHz, 80 W, 45-sec repeats	1.8 × 2.0 cm
Liang et al, 2005 (56)	288	PMWA	2.45 GHz, 16-gauge, 60 W for 300 sec, repeats	Mean 3.8 cm ± 1.6
Xu et al, 2005 (57)	137	PRFA (<i>n</i> = 84), PMWA (<i>n</i> = 53)	RF, 14-gauge expandable 10 times, 290 KHz, 200 W, 10 min; MWA, 2.45 GHz, 16-gauge, 60 W, 5 min	Mean 2.6 cm ± 1.2
Dong et al, 2003 (52)	234	PMWA vs PRFA	2.45 GHz, 10–80 W, 16-gauge, 5–30 min	Mean 4.1 cm ± 1.9
Shibata et al, 2002 (46)	72	PMWA (<i>n</i> = 36), PRFA (<i>n</i> = 36)	MWA, 2.45 GHz, 70 W, 16-gauge, 60-sec repeats; RF, 460 KHz, 30–60 W, 15-gauge expandable, 15 min	MWA, 2.2 cm; RF, 2.3 cm
Shibata et al, 2000 (66)	30	PMWA (<i>n</i> = 14), resection (<i>n</i> = 16)	2.45 GHz, 60–100 W, 16-gauge, 2–20 min	MWA, mean 2.7 cm ± 1.1; resection, mean 3.4 cm ± 1.7 (CRC)
Seki et al, 1999 (65)	15	PMWA	80 W, 14–15-gauge, variable time, repeats	Solitary lesions ≤ 3 cm (CRC)

Note.—CRC = colorectal cancer metastatic disease; HCC = hepatocellular carcinoma; LMWA = laparoscopic microwave ablation; NA = not applicable; OMWA = open microwave ablation; PMWA = percutaneous microwave ablation; PRFA = percutaneous RF ablation; TMWA = thorascopic-assisted microwave ablation.

enables production of larger ablation zones in the kidney, which has already been demonstrated in animal studies (70–72). Laeseke et al (70) compared microwave ablation with high-power triaxial antennas versus RF ablation with similarly sized internally cooled electrodes in an in vivo porcine kidney model. Microwave ablation created significantly larger ablation zones with higher tissue temperatures (123°C vs 100°C).

Early clinical studies have shown that microwave may be effective for the treatment of renal cell carcinoma. Clark et al (73) operatively treated 10 patients with large renal tumors (3.9–13 cm in diameter) with as many as three 13-

gauge microwave antennas powered at 60 W for 10 minutes before radical nephrectomy. Pathologic examination with viability stains demonstrated an average ablation zone volume of 105 cm³ (5.7 × 4.7 × 3.8 cm) when using a three-probe array. An important additional finding was that there was uniform cell death in the ablation zone (73). Liang et al (74) treated 12 patients with renal cell carcinomas (1.3–3.8 cm in diameter) with microwave and found complete ablation in a single session with no residual or recurrent tumor at a median follow-up of 11 months. Although these preliminary results are promising, further clinical experience in the kidney is needed.

Lung

In the lung, less invasive treatment modalities, particularly thermal ablation, have been increasing in popularity for medically unresectable lung tumors, both as a primary treatment and as an adjuvant therapy to external radiation (75–78). However, aerated lung has low electrical conductivity and poor thermal conduction, limiting the effectiveness of RF energy (79,80). Microwaves may offer significant advantages in the lung. The low conductivity and high impedance of aerated lung does not degrade the volume heating of microwaves. In fact, the lower permittivity and conductivity inherent in lung may allow deeper

Ablation Zone Size	Follow-up (mo)	Local Recurrence	Survival
—	Mean 22 ± 18.5 mo	MWA, 17%; RF, 26.4%	MWA median, 30.3 mo; RF median, 19 mo
—	Mean 24.6 ± 18.5 mo; median 21 mo, range 1–68 mo	PMWA, 15.1%; TMWA, 1.1%	HCC 3 y PMWA 89.9%; TMWA 89.9%; CRC 3 y PMWA 5%; TMWA 6.6%; other metastases 3 y PMWA 39%; TMWA 48%
Mean single volume, 10 mL; mean cluster volume, 50.5 mL	Mean 19 mo	2.7% (regional recurrence, 43%)	Survival 47% at 19 mo (all tumor types)
Single straight 16.7 cm ³ ; triple straight 51.7 cm ³ ; triple loop 54.3 cm ³	Resect	NA	NA
Mean diameter 5.5 cm, volume 50.8 cm ³	Resect	NA	NA
	MWA, 25.1 mo ± 12.7; RF, 24.8 mo ± 14.6	MWA, 11.8%; RF, 20.9%	MWA, 1y 81.6%, 2 y 61.2%, 3 y 50.5%, 4 y 36.8%; RF, 1 y 71.7%, 2 y 47.2%, 3 y 37.6%, 4 y 24.2%
Mean area 42.7 mm ²	Mean 31 mo ± 16	12%	1 y, 97%; 3 y, 90%; 5 y, 48%
—	Mean 31.4 mo ± 20.4	8%	Single nodule, 1 y 96%, 3 y 78%, 5 y 60%; multiple nodules, 1 y 91%, 3 y 61%, 5 y 28%
—	Mean 24 mo ± 1.5	—	Overall survival thermal ablation, 1 y 73.9%, 3 y 42.8%, 5 y 20.1%
—	27.9 mo ± 18.4	7.2%	1 y, 92.7%; 2 y, 81.6%; 3 y, 72.8%; 4 y, 66.4%; 5 y, 56/7%
—	Range 6–27 months	MWA, 10% at 1 y; 24% at 2 y; RF, 4% at, 1 y; 12% at 2 y	—
—	—	—	MWA, 1 y 71%, 2 y 57%, 3 y 14%; resection, 1 y 69%, 2 y 56%, 3 y 23%
—	9–37 mo	NA	67%

microwave penetration than is seen in the liver.

Microwaves have been shown to actively heat larger volumes of normal lung than comparable RF devices (3,12,23,81). Durick et al (23) showed that, with a microwave ablation system tuned for the lung, large circular ablation zones in porcine lungs in vivo could be obtained with a single antenna. Coagulation zones created with bronchial occlusion and multiple antennas were significantly larger than those seen without bronchial occlusion and with a single antenna, respectively (Table 3) (12,17,23,82). In a subsequent comparative trial, Brace et al (12) compared ablation zones obtained with a 17-gauge microwave antenna versus those ob-

tained with a 17-gauge RF electrode in an in vivo normal porcine lung model. Ablation zones generated with the use of microwaves were 25% larger in mean diameter and significantly more circular, and developed faster than those achieved with RF energy (Fig 6, Table 3) (12).

Note that none of these experimental protocols used a lung tumor model, and the properties of lung tumors may be more similar to those of solid organs. However, early clinical studies suggest that tumor heating with microwave energy is also effective (3,4,17,82). In the largest study to date, Wolf et al (17) percutaneously treated 82 lung masses (primary lung tumors and metastatic disease) in 50 patients. Local control rate (based on serial follow-up imaging

with computed tomography [CT] and positron emission tomography) at 1 year was 67%, with a mean time to first recurrence of 16.2 months (Table 3). Although preliminary data suggest that microwave ablation is safe and effective in the lung, further study is needed.

Bone

Bone has low conductivity (ie, high impedance) and thermal conduction, which serves to limit the effectiveness of RF ablation. Microwave is relatively insensitive to high impedance and has a deeper penetration profile. Both these characteristics may represent a relative advantage for microwave ablation in this setting. As a result of the relative

Table 3
Preclinical and Clinical Microwave Trials in Lung (12,17,23,82)

Study	No. of Pts.	Treatment	System Parameters	Tumor Size
Brace et al, 2009 (12)	18	PMWA, 9 ablations; PRFA, 9 ablations	MWA, 2.45 GHz, 125 W, 17-gauge, 10 min; RF, 200 W, 17-gauge, 10 min	NA (in vivo porcine)
Wolf et al, 2008 (17)	50	PMWA	915 MHz, 45–60 W, 14.5-gauge, 5–10 min	Mean diameter 3.5 cm $\pm$ 1.6
He et al, 2006 (82)	12	PMWA	2.45 GHz, 30–100 W, 14-gauge, variable time	Range 3–6 cm (lung, metastases)
Durick et al, 2005 (23)	24	OMWA	2.45 GHz, 55 W, 17-gauge, 10 min	NA

Note.—NA = not applicable; NR = not reported; OMWA = open microwave ablation; PMWA = percutaneous microwave ablation; PRFA = percutaneous RF ablation.

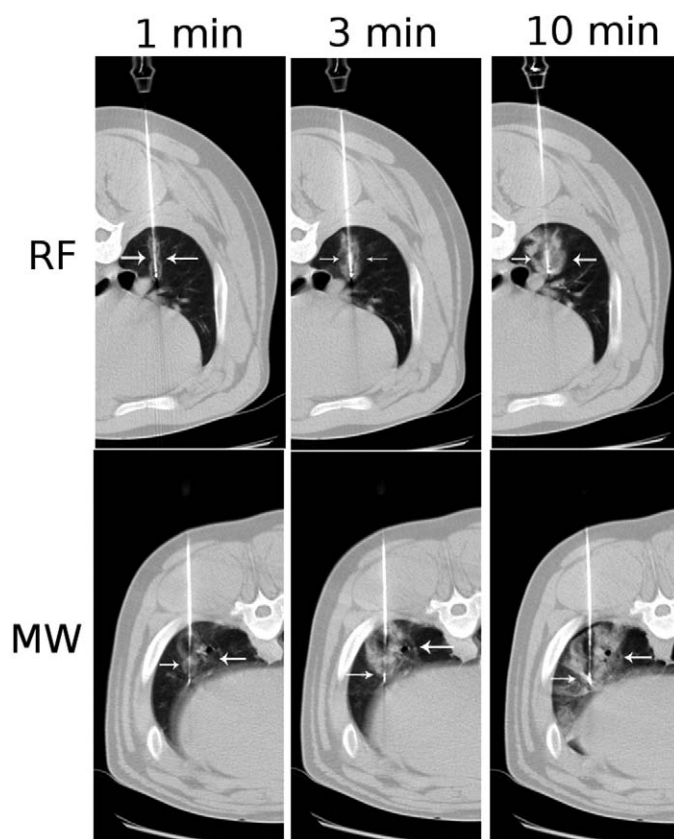


Figure 6. Sequential CT images obtained during RF versus MW ablation in porcine lungs in vivo. Note the larger ablation zones (arrows) at each time point in the microwave panel. Figure reprinted with permission from Brace et al (12).

permittivity of bone, microwave may be less affected by tissue heating and desiccation, providing deeper penetration and more effective heating. Two different applications for ablation in bone currently exist: treatment of painful osteoid osteomas and treatment of other bone tumors, particularly painful metastases. There is very little experience with treat-

ment of osteoid osteomas with microwaves, perhaps because a microwave system capable of producing short ablation zones (approximately 1 cm) is not yet available. Currently available antennas produce zones of ablation as long as 5–7 cm. Therefore, microwaves may not currently have clinical application in the treatment of osteoid osteomas given the

risk of nontarget thermal damage. Early clinical use of microwave in other bone tumors/metastatic disease has been promising. For example, Fan et al (83) treated 213 patients with malignant bone tumors of the extremities and pelvis with combined operative intervention and microwave ablation therapy in a limb-sparing procedure, with a survival rate of 74% at an average follow-up of 49 months. Further study is needed in both areas to confirm the theoretical advantages in bone, as only a few cases have been published in the literature to date (4,13–15,83).

DISADVANTAGES OF MICROWAVE

Microwave power is inherently more difficult to generate and deliver safely and efficiently to the tissue compared with RF. This is primarily because microwave energy must be carried in coaxial cables, which are larger in diameter, more cumbersome, and more prone to heating than the simple wires used in RF ablation. Decreased cable surface area leads to more power loss and increased cable heating. As one of the primary advantages of microwave is the ability to deliver large amounts of power, the technical hurdles to distribute this power to tissues without significant cable and shaft heating must be overcome before this advantage can be fully realized. A robust active shaft cooling mechanism can mitigate many of these risks and is imperative to high power delivery (Fig 7). A clinical study comparing cooled with noncooled antennas in a cohort of 1,136 patients (84) showed that use of the cooled-shaft antenna led to fewer treatment sessions

Ablation Zone Size	Follow-up	Local Recurrence	Survival
MWA, 3.8 cm $\pm$ 0.3 max diameter; RF, 3.3 cm $\pm$ 0.4	NA	NA	NA
NR	Mean 10 mo	26% residual, 22% recurrence	1 y, 83%; 2 y, 55%; 3 y, 45%
NR	Average 20 mo	Tumors decreased in size	58% at mean follow-up
Mean diameter, 3.4 cm (single), 4.8 cm (multiple)	NA	NA	NA

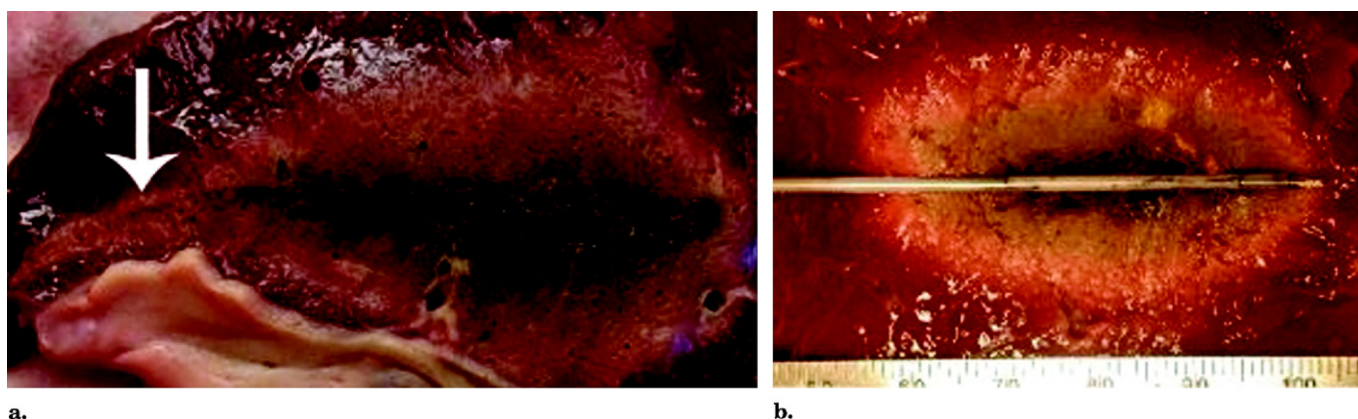


Figure 7. (a) Noncooled antennas can cause proximal tissue heating, creating an unwanted “tail” of ablation in the more superficial tissues (arrow), as seen in this photograph of a hepatic microwave ablation zone. However, cooling eliminates this unwanted shaft heating and ablation tail (b), with a hepatic ablation zone generated with an identical power and time but with the addition of water cooling of the antenna shaft.

and fewer major procedural complications.

Currently available microwave systems continue to face technical limitations, and this has prevented the potential of microwave from being realized in the clinic to date. Major limitations include underpowered systems, shaft heating, large diameter probes, and long and relatively thin ablation zones with limited clinical application (especially in small bone lesions such as osteoid osteomas and solid organ surface lesions). Similarly, there is still some unpredictability as to the size and shape of the zone of ablation, which may be related to technical factors (Figs 8, 9).

CURRENTLY AVAILABLE MICROWAVE SYSTEMS

In the United States, only one Food and Drug Administration–cleared microwave system is widely available for

commercial use (ValleyLab/Covidien, Boulder, Colorado). The Evident system is a 13-gauge water-cooled dipole antenna with a 915-MHz generator and a maximum recommended output of 45 W (33,85). Preclinical testing was first performed by Wright et al (20), who demonstrated that simultaneous multiprobe ablations were nearly six times larger in volume than single-probe ablations. Oshima et al (22) also explored the use of multiple applicators with this system and found that, when spaced 2 cm or less apart, spherical ablation zones were maintained, with significant increase in size of the zone compared with those created with a single antenna. Iannitti et al (54) used this system in a phase II clinical trial to treat 87 patients with hepatic tumors. Mean follow-up to 19 months demonstrated local tumor progression in 2.7% and regional recurrence/new foci of disease in

the target organ in 43%. Martin et al (86) also used this system for a prospective phase II study of microwave ablation of hepatic malignancies (hepatocellular carcinoma and metastatic disease, most treated operatively) in 100 patients. At a median follow-up of 36 months, 5% of patients had incomplete ablation (seen at initial 2-week follow-up imaging), 2% had local tumor progression, and 37% exhibited intrahepatic recurrence at nonablated sites. These early results are promising, but the system needs more extensive in vivo characterization and validation in larger clinical trials before its full potential will be known (Figs 8, 9).

Other systems that have recently received Food and Drug Administration 510(k) premarketing clearance or are currently in development include the MicrothermX-100 (BSD Medical; Salt Lake City, Utah), which uses a 915-MHz generator and multiple 14-gauge appli-

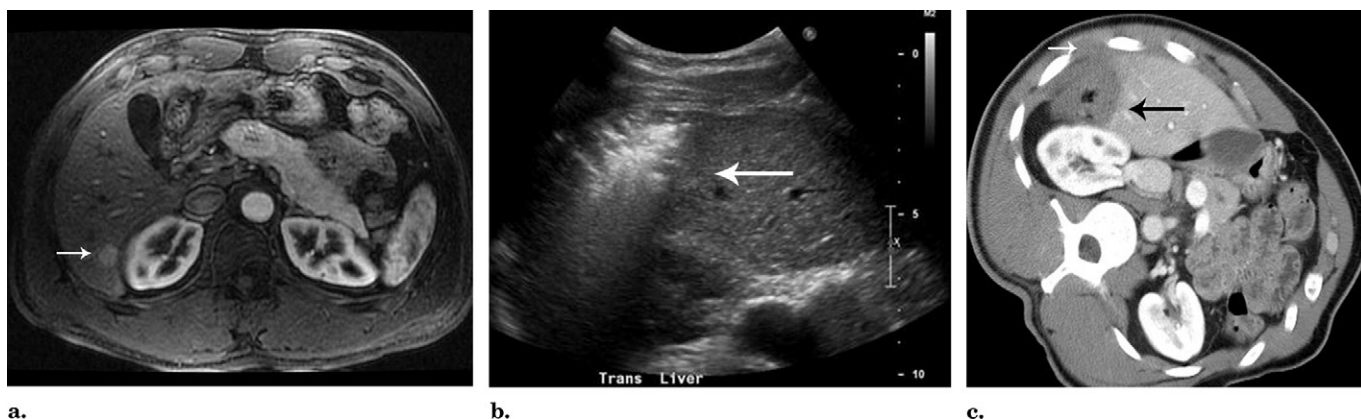


Figure 8. Microwave ablation of small hepatocellular carcinoma with the Covidien Evident system. CT image before ablation (a) demonstrates a small hepatocellular carcinoma (arrow) in a patient with hepatitis C and relatively mild cirrhosis. Two probes were placed and ablation was performed, with a large area of gas bubbles forming (arrow) on the periprocedural ultrasound image (b). These probes, run at 45 W for 10 minutes, generated a larger than expected ablation zone (large arrow) that extended into the body wall (small white arrow) seen on postprocedure CT (c).

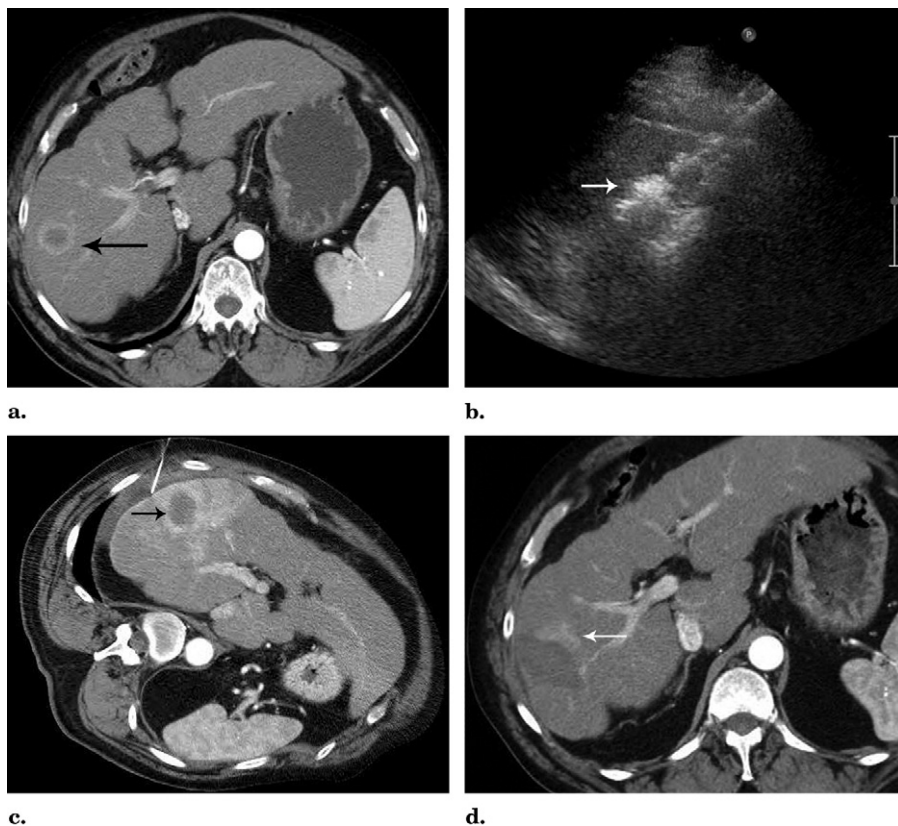


Figure 9. Microwave ablation of liver lesion was performed with use of a Covidien Evident system. Preprocedure CT (a) demonstrates a small hepatocellular carcinoma (arrow) in a mildly cirrhotic liver. Two probes (b) were placed into the lesion and run at 45 W for 10 minutes, similar to the case in Fig 8, above. However, a relatively small area of hyperechoic change was identified on the periprocedural US (arrow, b) and the immediate postprocedure CT image (arrow, c) demonstrates a much smaller ablation zone with residual tumor seen along the anterior and medial margins at 1-month follow-up (arrow, d). The variation in result between these two cases may be related to increased sensitivity of microwaves to probe placement and phasing.

catators with individual antenna control; the AveCure system (MedWaves; San Diego, California), which uses a 915-MHz generator and 12–16-gauge antennas with temperature feedback rather than cooling; and the Certus 140 (Neuwave Medical, Madison, Wisconsin), a 2.45-GHz system under development that supports as many as three actively gas-cooled 17-gauge triaxial antennas.

The Acculis microwave tissue ablation system (Aculis, Denmead, United Kingdom), available in Europe, consists of a 2.45-GHz generator, with power output settings ranging from 30 to 100 W, and a percutaneous 15-gauge cooled antenna. The Amica microwave system (HS Medical, Boca Raton, Florida) uses a 2.45-GHz generator, with power output to 100 W (60 W recommended output) through 14-gauge and 17-gauge water-cooled applicators (Fig 10).

To date, microwave ablation has been used most widely in Japan and China, where multiple systems have been developed. Most of these systems use 2.45-GHz generators with monopole, dipole, or slotted coaxial antennas. For example, the Forsea system (Qinghai Microwave Electronic Institute, Nanjing, China) uses a 10–150-W, 2.45-GHz generator and 14-gauge water-cooled antenna. Kang-Yu Medical manufactures 915-MHz and 2.45-GHz generator systems (KY2000-915 and KY2000-2450, respectively) with a water-cooled 15-gauge antenna. Both generators can produce 1–100 W. The Microtaze device (Nippon Shoji, Osaka, Japan) also uses a 2.45-GHz generator

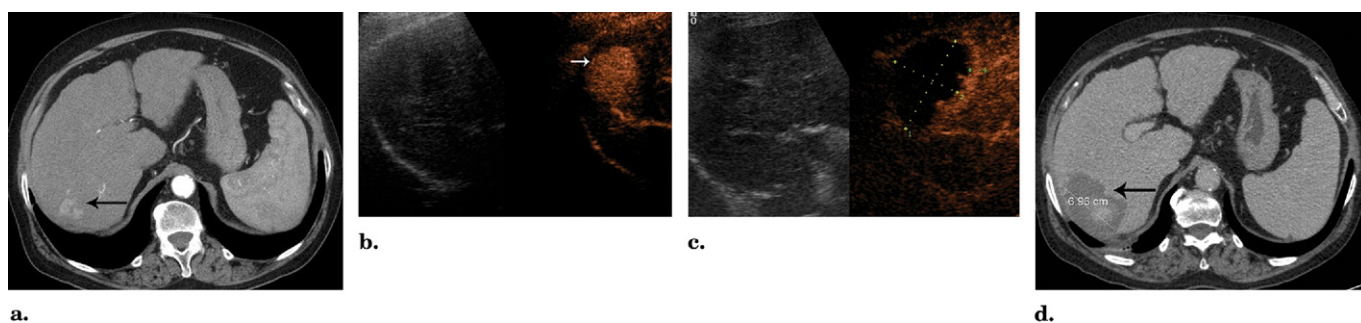


Figure 10. Microwave ablation performed with an HS Amica system. Preprocedure CT demonstrates nodular arterially enhancing lesion in the posterior right hepatic lobe (arrow, a), demonstrated again on contrast-enhanced preprocedure US (arrow) (b). Three sequential placements were performed with an antenna powered at 60 W for 15 minutes each. This produced an ablation zone nearly 7 cm in maximum diameter, as seen on postprocedure ultrasound (c) and CT (arrow, d). Case courtesy of M. Franca Meloni and Anita Andreano, University of Milano–Bicocca.

Table 4
Currently Available Microwave Systems and Systems in Development

Device Manufacturer	Antenna Design	Antenna Size (gauge)	Cooling Type	Power (W)	Frequency	Multiple Applicators
Valleylab/Covidien	Straight	13	Water	45	915 MHz	No
BSD Medical	Straight	14	Water	60	915 MHz	Yes
MedWaves	Straight	12–16	None	32 (modulated)	915 MHz	No
Neuwave	Triaxial, straight	17	Gas	140	2.45 GHz	Yes (3)
Acculis	Straight	15	Water	100	2.45 GHz	No
HS Amica	Straight	14, 17	Water	100	2.45 GHz	Yes (2)
Forsea	Straight	14	Water	150	2.45 GHz	Yes (2)
Kang-you Medical	Straight	15	Water	100	915 MHz, 2.45 GHz	—
Microtaze	Straight	16	Water	70	2.45 GHz	—

with a 16-gauge cooled-shaft antenna with outputs between 60 and 70 W. For a summary of available systems, see [Table 4](#). Given the substantial limitations in commercially available devices to date, commercial and academic development is ongoing.

CONCLUSIONS

In summary, microwave tumor ablation is an exciting technique with many theoretical advantages over currently existing ablation systems, including generation of larger ablation zones in shorter times with very high temperatures and less susceptibility to heat-sink effects. In addition, microwaves have an advantage in high-impedance tissues such as lung and bone that have been problematic for other ablation modalities. Preclinical and early clinical studies have begun to confirm these advantages, but further study and continued development of more robust clinical systems is still needed.

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